Star Schema: Online Shop (shop.db)

Source: customers, products, orders, order_items → 1 fact table + 4 dimensions PK primary key FK foreign key NK natural key 1 / N cardinality

dim_customer	SCD Type 2
customer_key	PK
customer_id	NK
name	
email	
city	
signup_date	
valid_from	
valid_to	
is_current	

dim_product	SCD Type 2
product_key	PK
product_id	NK
name	
category	
list_price	
valid_from	
valid_to	
is_current	

fact_order_items	
order_item_key	PK
order_id (degenerate dim)	
customer_key	FK
product_key	FK
date_key	FK
status_key	FK
quantity	
unit_price	
line_amount = qty x price	

dim_date	SCD Type 0
date_key (YYYYMMDD)	PK
full_date	
year	
quarter	
month	
month_name	
day	
day_of_week	
is_weekend	

dim_order_status	SCD Type 1
status_key	PK
status_name	
is_revenue	

GRAIN

One row = one product line
of one order (one order_items row)

Measures: quantity, unit_price, line_amount

dim_customer	**Type 2**	city/email can change; old orders must keep the city the customer had at purchase time.
dim_product	**Type 2**	category and list price change over time; history is needed for price/category analysis.
dim_date	**Type 0**	a calendar date never changes.
dim_order_status	**Type 1**	tiny lookup; a label fix just overwrites, no history needed.